

Do Television Series Decline Over Their Run?

Evidence from 192,720 IMDb Episode Ratings

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Abstract

The belief that television series decline over their run is widely held and rarely measured. Using the complete IMDb episode-rating corpus, we construct a panel of 192,720 rated episodes from 3,234 series and estimate a per-show quality trajectory as the slope of a vote-weighted least-squares fit of episode rating on normalised episode order. We find the folk belief is false for the typical series: only 16.9% of shows decline by at least half a rating point across their entire run, 57.4% are essentially flat, and 25.7% clearly rise. A coupling-free test—regressing each show’s second-half mean on its separately measured first-half mean—yields a correlation of +0.793 and a slope of 0.886 against a noise-only expectation of 0.983, implying a real but negligible extra decline of 0.07 rating points for the strongest starters. Season finales exceed the remainder of their own season 72.6% of the time, and series finales exceed their final season 71.6% of the time, so the moment the belief expects collapse is typically a local maximum. Among ended series the final season is weaker than the rest of the run in 50.6% of cases with a median change of -0.006 , and only 12.6% lose half a point or more; this conclusion is insensitive to the threshold chosen. Decline is concentrated rather than general: show length (-0.026 per season, $t = -6.8$) and premiere year ($+0.048$ per decade, $t = +4.8$) each survive mutual adjustment, whereas genre is largely uninformative once age and length are controlled. The model explains only 6.0% of between-show variance, so these are reliable but small effects. We argue the persistence of the belief is consistent with availability bias: the minority of genuine collapses is disproportionately composed of highly-viewed series.

Keywords: television ratings, quality trajectories, regression to the mean, availability bias, IMDb

1 Introduction

That television series deteriorate over time is close to received wisdom. The claim is specific enough to be testable—it asserts that a show’s quality declines monotonically or near-monotonically with time on air—yet it is usually supported by enumeration of memorable cases rather than by measurement.

This report tests the claim directly. We ask five questions:

1. What is the distribution of per-show quality trajectories?
2. Is a show’s later quality predictable from its earlier quality, and does high early quality decay?
3. Do finales—season and series—underperform, as the belief implies?

4. Is there a “final-season curse”?
5. If decline exists, which show characteristics predict it?

Our contribution is not a new method but a disciplined application of standard ones to a question usually argued anecdotally, with explicit attention to the artefacts that make naive versions of this analysis produce the opposite answer. Section 5 documents four such artefacts, one of which produced an apparently strong effect that survived neither reformulation nor replication.

2 Data

2.1 Source

All ratings come from the IMDb public datasets [1], downloaded on **22 July 2026**. IMDb republishes these files daily; counts drift as new episodes accumulate ratings and borderline series cross inclusion thresholds, so the exact figures below are reproducible only against the committed snapshot. Three tables are used: `title.episode` (mapping each episode to its parent series, season and episode number), `title.ratings` (mean rating and vote count per title) and `title.basics` (title type, primary title, start year, end year, genres).

2.2 Construction

Episodes with a missing season or episode number are dropped, since ordering within a series is required. The episode table is inner-joined to ratings—so only episodes carrying at least one vote survive—and then to the series rows of `title.basics` restricted to `tvSeries` and `tvMiniSeries`. Episodes are ordered by season then episode number, and a within-show sequential index is assigned. Series with an empty end year are flagged as ongoing rather than dropped. The result is one row per rated episode.

2.3 Inclusion criteria

Six filters define the analysis population (Table 1).

Table 1: Inclusion criteria applied to the joined episode table.

Criterion	Threshold	Rationale
Title type	<code>tvSeries</code> , <code>tvMiniSeries</code>	excludes films and specials
Rated episodes	≥ 13	a trajectory needs enough observations
Seasons	≥ 2	decline across seasons is otherwise undefined
Mean votes per episode	≥ 50	limits noise from thinly-rated series
Genre exclusions	not <code>Game-Show</code> , <code>Reality-TV</code> , <code>Talk-Show</code> , <code>News</code>	non-narrative formats have no season arc to decline
Documentary cap	kept only if < 100 episodes	separates season-based series from magazine strands

The documentary rule is platform-independent, since the data carry no channel or platform field.

2.4 Sample

Table 2 describes the resulting panel.

Table 2: The analysis sample after all six filters.

Property	Value
Series	3,234
Rated episodes	192,720
Total votes represented	160,926,754
Premiere years	1949–2026
Ongoing at snapshot	524 (16.2%)
Episodes per series	median 36 (IQR 22–69, max 1,234)
Seasons per series	median 3 (IQR 2–5, max 70)
Mean rating per series	median 7.72 (IQR 7.39–8.03)
Votes per episode	median 190

The sample is right-skewed in length: 485 series exceed 100 episodes and 127 exceed 200. These were retained rather than trimmed; the trajectory metric normalises episode order to $[0, 1]$ so that run length cannot mechanically inflate a slope, and length is analysed as an explanatory variable in its own right (Section 4.5).

3 Methods

3.1 Per-show trajectory

For each series, let $x_i \in [0, 1]$ be the normalised position of episode i in the run and y_i its mean rating. We fit

$$y_i = a + b x_i + \varepsilon_i$$

by weighted least squares with weights $w_i = \sqrt{v_i}$, where v_i is the episode’s vote count.

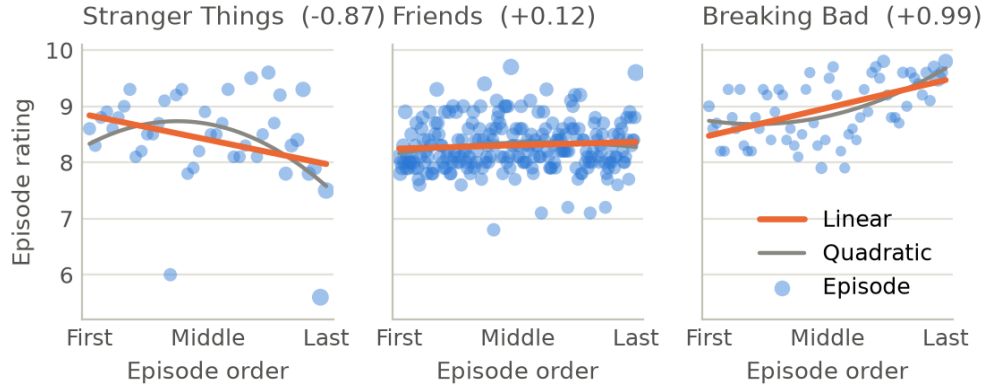


Figure 1: Three fitted trajectories, one from each outcome band defined below. Points are episode means sized by vote count; the straight line is the estimate $\hat{b}$, the grey curve the quadratic of Section 3.3.

The estimate $\hat{b}$ is the show’s *trend*: the rating points gained or lost between first and last episode. A show is classified as *clearly declining* if $\hat{b} \leq -0.5$, *clearly rising* if $\hat{b} \geq +0.5$, and *essentially flat* otherwise. The half-point cut is a perceptibility convention, not a statistical one; Section 5 reports its sensitivity.

Choice of weights. Vote counts vary substantially within a series (median max/min ratio 3.5, exceeding 10 for a tenth of series), so an unweighted fit lets a 51-vote episode influence the trend as much as a 500,000-vote one. Weighting by v_i directly is optimal under an i.i.d.-votes noise model, but IMDb vote counts violate it: notorious episodes attract votes for reasons unrelated to measurement precision. In *Game of Thrones*, four of the six most-voted episodes belong to the final season and all are rated far below the series mean; linear weighting nearly doubles the fitted slope relative to an unweighted fit (-2.206 versus -1.024), while $\sqrt{v_i}$ gives -1.546 . Log-weighting was also examined and is nearly indistinguishable from unweighted in practice. We therefore use $\sqrt{v_i}$ throughout.

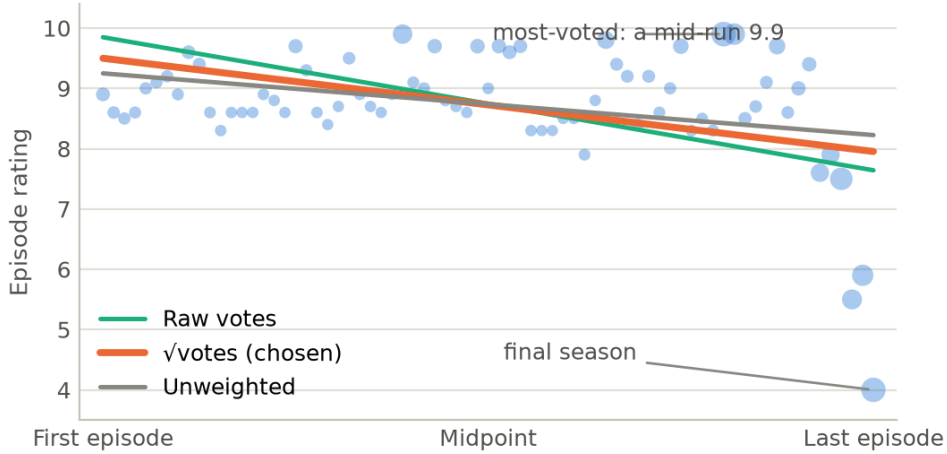


Figure 2: The same 73 episodes under three weighting schemes. Raw vote weighting lets the heavily-voted final-season cluster set the trend for the whole run; the single most-voted episode is in fact a mid-run 9.9.

3.2 Final-season effect

The slope in Section 3.1 answers “does this series decline at a roughly constant rate”, which is *not* the final-season question: a straight line fitted across all episodes is anchored by the majority and understates a late, sharp drop. For *Game of Thrones* the true season-8 mean is 6.40 against 8.95 for seasons 1–7, yet the fitted line predicts 7.6–8.3 in that range under every weighting scheme. We therefore define, for ended series only,

$$\Delta = \bar{y}_{\text{final season}} - \bar{y}_{\text{all earlier episodes}},$$

and use Δ for all final-season claims. A series is termed *clearly cursed* if $\Delta \leq -0.5$.

3.3 Trajectory shape

A weighted quadratic $y = a + bx + cx^2$ is fitted per series with the same weights. Where the vertex $-b/2c$ lies inside $[0.15, 0.85]$ and $|c| > 0.05$, the series is labelled **jumped-the-shark** ($c < 0$: interior maximum) or **found-itself** ($c > 0$: interior minimum); otherwise **straight line**, meaning the fitted curve has no turning point inside the observed run and is monotone over it. Note this last class is not “flat”: it contains steep monotone decliners and risers alike. Volatility is the weighted RMSE of the linear fit’s residuals.

3.4 Peak location

For each series we take the argmax of season means, reported both as an absolute season number and as a normalised rank $(\text{rank} - 1)/(n_{\text{seasons}} - 1)$. Because the sample is dominated by short series, aggregate statements about the peak are reported stratified by season count.

3.5 Cross-sectional model

To separate era, length and genre we estimate by OLS

$$\hat{b}_j = \beta_0 + \beta_1 \text{year}_j + \beta_2 \text{seasons}_j + \beta_3 \text{ongoing}_j + \sum_g \gamma_g \mathbf{1}[g \in \text{genres}_j] + u_j$$

over all $n = 3,234$ series, with premiere year centred and genres entered as 16 non-exclusive indicators. Classical standard errors are reported; given the sample size and the magnitude of the leading t -statistics, robust alternatives would not change any conclusion drawn here.

3.6 Software and figures

The dataset is built and analysed with NumPy [7] and pandas [8]; figures are drawn with Matplotlib [9]. Each figure recomputes its own inputs from the processed table and prints the values it annotates, so a plate cannot drift away from the number quoted beside it. Figure colours were checked rather than chosen by eye: the palette was validated for colour-vision deficiency using the Machado *et al.* [10] simulation at full severity, which rejected one pair of hues that had been used together.

4 Results

4.1 The distribution of trajectories

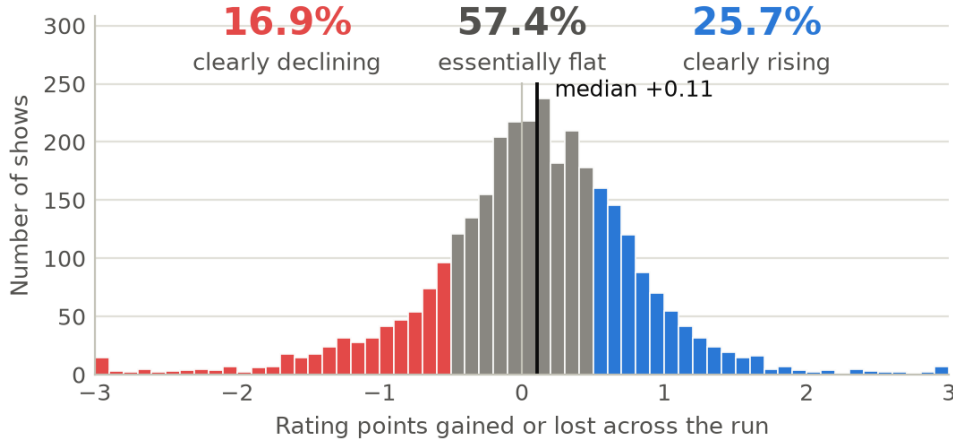


Figure 3: Distribution of the per-show trend $\hat{b}$ across all 3,234 series. Red marks clear decline ($\hat{b} \leq -0.5$), blue clear rise, grey the flat band between them; the black rule is the median. The axis is clipped to ± 3 points, which places 18 series (0.6%) outside it.

The trend distribution is centred slightly above zero (mean $+0.073$, median $+0.109$, SD 0.778 , IQR -0.296 to $+0.515$). By sign, 57.4% of series trend upward and 42.6%

downward. By magnitude, **16.9% decline clearly**, **57.4% are essentially flat**, and **25.7% rise clearly**. Under the folk belief we would expect a distribution massed below zero; we observe a distribution massed *at* zero with a mild positive tilt.

4.2 Stability across a series' run

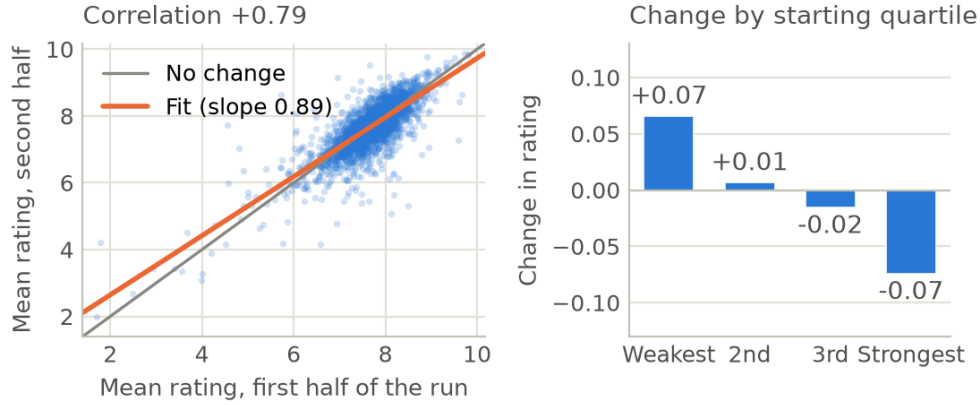


Figure 4: Each series' first-half mean against its separately measured second-half mean, all 3,234 series. The grey line is equality, the orange line the fitted relationship. Right: mean change from first half to second, by quartile of first-half rating.

Splitting each series into disjoint halves, first-half and second-half means correlate **+0.793**. Regressing second on first yields a slope of **0.886**, below unity and therefore consistent with some regression to the mean. The relevant benchmark is not 1 but the reliability of a half-mean: an odd/even split-half correlation with the Spearman–Brown correction [2, 3] gives **0.983**, which is what a pure “stable quality plus measurement noise” process would produce. The observed 0.886 falls modestly below this, so a genuine extra decline of strong starters exists—but it is very small. In levels, top-quartile starters (first-half mean 8.32) lose **0.07** rating points in their second half, and bottom-quartile starters (6.94) gain 0.07.

4.3 Finale structure

Restricting to seasons of at least four rated episodes ($n = 12,931$ seasons), the season finale beats the rest of its own season by **+0.253** rating points on average and does so 72.6% of the time; the season premiere is slightly below its own season (-0.040 , above baseline 43.3%); and the series finale beats its final-season body by **+0.256**, 71.6% of the time ($n = 3,082$ series). Figure 5 plots all three.

Finales are systematically the strongest episode of their season, premieres are mildly below average, and the asymmetry is large. Only 9.4% of series finales fall 0.5 points or more below their final season—the memorable “finale flop” is a one-in-ten event.

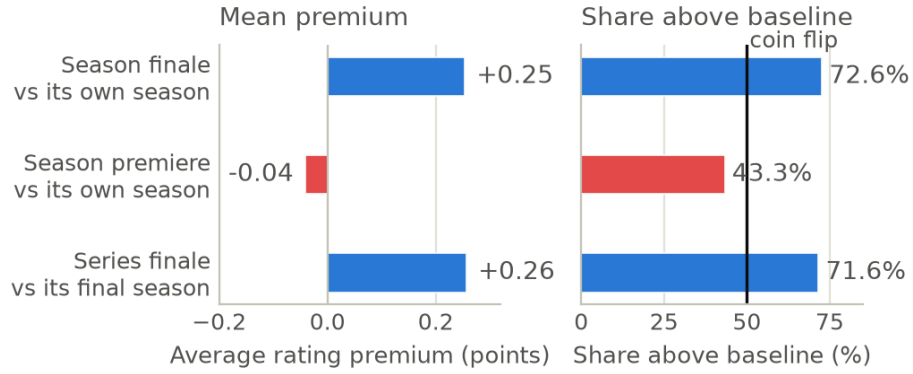


Figure 5: Finale and premiere premia relative to the rest of the season. Left: mean premium in rating points. Right: share of cases beating the comparison group, against the 50% line. Seasons of at least four rated episodes, $n = 12,931$ seasons and 3,082 series finales.

4.4 The final-season effect

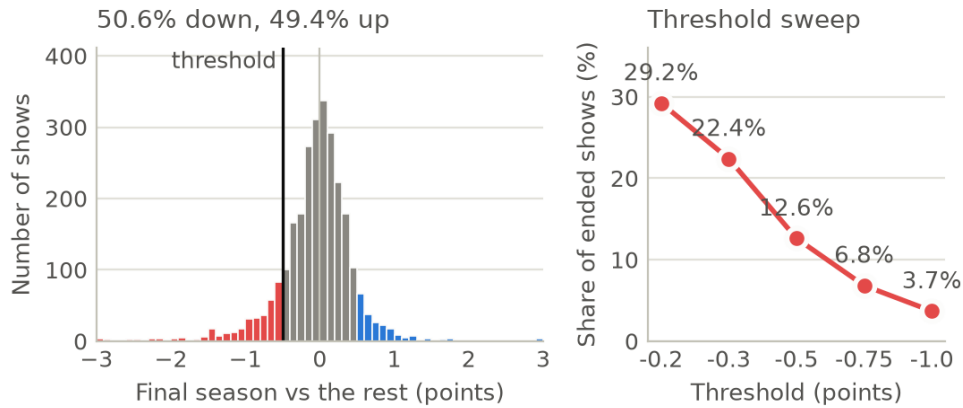


Figure 6: Distribution of Δ for the 2,710 ended series, with a sensitivity sweep over the “cursed” threshold. The black rule marks $\Delta = -0.5$, which 12.6% of ended series fall below. The axis is clipped to ± 3 points, leaving 6 series outside it.

Across 2,710 ended series, 50.6% have a weaker final season and 49.4% a stronger one, with median $\Delta = -0.006$ and mean -0.049 . **12.6%** are clearly cursed at the -0.5 cut. The cursed minority is dominated by well-known series (*House of Cards* -4.24 , *Game of Thrones* -2.55 , *Master of None* -2.54 , *The Promised Neverland* -2.28), which we read as the mechanism sustaining the belief: the observable instances are disproportionately the famous ones, so the cases available to recall are not a fair sample of the cases that exist [6]. This interpretation is not established by our data; it is consistent with them.

Δ and $\hat{b}$ correlate $+0.80$, so roughly a third of their variation is independent and they are not interchangeable. Notably, the pure “flat then cliff” case is essentially absent: all 32 series with $\Delta \leq -1.0$ and at least 30,000 votes also have a negative overall slope, which is mechanical—a late crash necessarily tilts a fitted line downward.

4.5 Era, length and genre

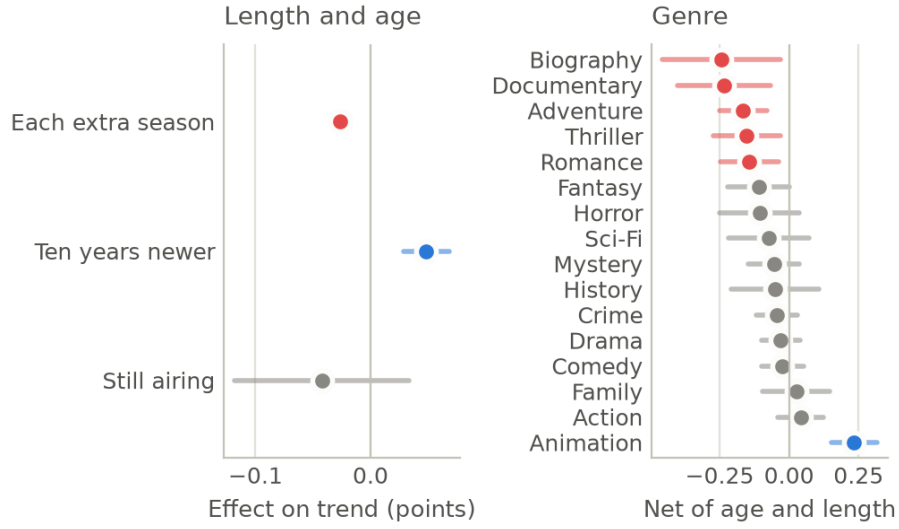


Figure 7: OLS coefficients with 95% confidence intervals, $n = 3,234$. A coloured dot marks $|t| > 2$; grey marks an estimate not distinguishable from zero. Era is shown per decade for comparability with the per-season length effect; the model itself is per year. Full table in Appendix A.

Raw cross-sectional cuts are heavily confounded: median seasons per series falls from 5 in the 1950s to 2 in the 2020s while the ongoing share rises from 3% to 41%. After mutual adjustment (full table in Appendix A):

- **Length:** -0.0261 per season ($t = -6.78$). Ended 2–3-season series average $+0.217$; ended 6+-season series -0.120 , with 60.8% of 6+-season series declining.
- **Era:** $+0.0048$ per year, i.e. $+0.048$ per decade ($t = +4.81$). The effect survives restriction to ended series and to fixed length bands, so it is not a length or ongoing-status artefact. 59.3% of pre-2000 series decline against 38.6% of later ones.
- **Genre:** only Animation ($+0.233$, $t = 5.57$), Documentary (-0.235), Biography (-0.244), Adventure (-0.165), Thriller (-0.153) and Romance (-0.143) reach significance. Drama, Comedy, Crime and Action—the bulk of the sample—are indistinguishable from zero.

The model explains 6.0% of between-show variance ($R^2 = 0.0603$, residual SD 0.756). The effects are estimated precisely but are small relative to the idiosyncratic differences between series; no combination of a show’s age, length and genre comes close to determining its trajectory.

4.6 Shape and peak location

The rise-then-fall arc is the single most common shape (**44.6%**), against 35.2% straight-line and 20.3% found-itself. Within the jumped-the-shark class the vertex sits at a median normalised position of 0.50, with only 6.2% of peaks in the 0.15–0.25 band, so the label describes a genuine interior maximum rather than a decline from

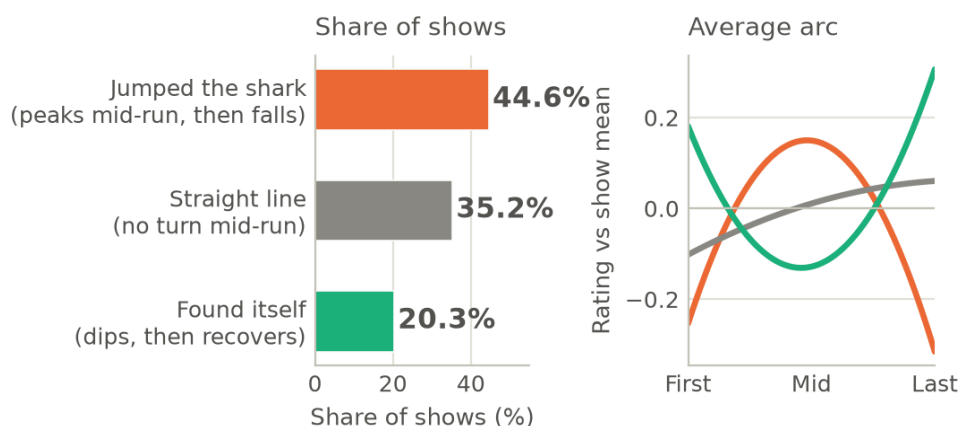


Figure 8: Share of series by trajectory shape, and the mean fitted arc of each class after centring on the series’ own mean rating. The straight-line class contains both rising and falling series, so its average is their net. Class shares move by less than one point across curvature cut-offs from 0 to 0.20; roughly a fifth of the found-itself class rests on thinly-voted late episodes (Section 5).

the outset. The arc is nonetheless shallow: median rise to the peak is 0.37 rating points and median fall from it 0.33. The classification is insensitive to the curvature cut-off—class shares move from 44.6/35.1/20.3 at a threshold of 0 to 43.7/36.6/19.8 at 0.20—because the binding condition is the location of the vertex, not the size of c .

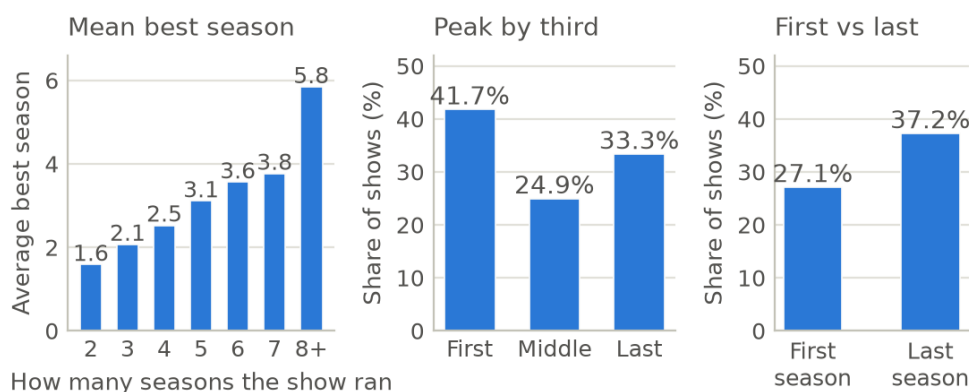


Figure 9: Location of the best season. Left: mean best season by run length, all 3,234 series. Centre: share of series peaking in each third of their run, restricted to the 915 series with at least five seasons, where thirds are meaningful. Right: how often the first and the last season is the single best, all series.

The mean best season rises with run length (1.6 at two seasons, 2.5 at four, 3.1 at five, 3.8 at seven, 5.8 at eight or more) while its normalised position remains near the middle (0.46–0.59 across all lengths). An earlier cut on three-season series alone suggested a “sophomore surge”; that is an aggregation artefact of a sample dominated by short series, in which season 2 *is* the middle. Importantly, the mean masks the distribution: among series with at least five seasons ($n = 915$), the peak falls in the first third 41.7% of the time, the middle third 24.9%, and the last third 33.3% —the middle is the *least* likely location, and the mean sits at 0.5 only because the ends

balance. Finally, the last season is the single best season 37.2% of the time against 27.1% for the first, contradicting the “first season is always best” claim in aggregate.

5 Robustness and threats to validity

Mathematical coupling. An early formulation correlated each show’s slope with its own fitted intercept, producing $r = -0.373$ and the apparent finding that high-quality series decline hardest. This is an artefact: both quantities are estimated from the same episodes and share noise, so the correlation is partly manufactured [4]. The disjoint-halves design in Section 4.2 removes the shared-noise channel entirely and reduces the effect to 0.07 rating points. We report the clean estimate and retain the biased one only as a documented negative result.

Metric–question mismatch. As shown in Section 3.2, a single linear slope cannot represent a flat-then-cliff trajectory and understates late crashes. Using the slope as a proxy for the final-season question would have understated *Game of Thrones*’ season-8 drop by roughly 1.2–1.9 rating points. All final-season claims therefore use the direct season contrast.

Threshold sensitivity. The clearly-cursed share is 29.2% at $\Delta \leq -0.2$, 22.4% at -0.3 , 12.6% at -0.5 , 6.8% at -0.75 and 3.7% at -1.0 : the conclusion that cursed final seasons are a minority holds at every reasonable cut. As a scale-free check we also computed Δ divided by the SD of the series’ own non-final-season episode ratings; 23% of ended series reach a “large” effect in Cohen’s conventional sense [5], and 98% of series flagged by the raw -0.5 rule also clear the large-effect bar, so the intuitive and principled criteria select nearly the same set.

Low-vote tails. Within a series, episode position correlates -0.74 with log vote count: later episodes are systematically less voted. Any late-leaning result is therefore on weaker ground than an early-leaning one. This matters most for the found-itself class: the median last-third episode carries only about 132 votes and 39.7% of found-itself series have a last third averaging under 100 votes. Re-fitting after dropping sub-100-vote episodes, 56.5% of found-itself series retain the label, 20.2% revert to decline or flat, and 23.4% become untestable. The label is not a weighting artefact (94.5% survive an unweighted refit) but roughly a fifth of it is a thin-data artefact, and we treat it accordingly.

Confounding. “Newer series rise”, “shorter series rise” and “older series decline” are not three findings: newer series are shorter. The specification in Section 3.5 holds each fixed, and both era and length survive. We additionally verified the era effect within ended series and within a fixed 2–3-season band, where it does not fade.

6 Limitations

Ratings are not quality. IMDb scores are contributed by self-selected voters, not a representative audience panel. Some low outliers reflect coordinated review-bombing rather than dispersed judgement; vote weighting dampens but does not remove this.

Ratings are a present-day snapshot. Each rating is the value standing today, not the reception recorded at broadcast. A series that ended in 2010 is scored in 2026 largely by viewers who arrived later and already knew its ending. The finale premium may therefore partly reflect retrospective softening. The dataset carries no time dimension with which to test this.

Causes are unobserved. The design measures where ratings move, never why. Off-screen shocks are indistinguishable from creative decline: *House of Cards* lost its lead actor amid scandal and had its final season cut from thirteen episodes to eight; *Top Gear* lost its presenting team; *Scrubs* became a different programme in its ninth season. All three appear in the data simply as “decline”.

Format leakage. The genre filters remove the main non-narrative categories, but sketch and review programmes tagged only as Comedy survive—most consequentially *Saturday Night Live* (1,010 episodes). A curated blacklist was rejected as unreproducible; the affected count is too small to move any reported figure, and we disclose it rather than patch it.

Survivorship in the vote threshold. Requiring at least 50 mean votes per episode excludes obscure series, so results generalise to series with a measurable audience rather than to all television.

7 Conclusion

The proposition that television series decline over their run is not supported for the typical series. Most series are close to flat across their entire run, more rise than fall, later quality is strongly predictable from earlier quality, and the extra decline of strong starters is an order of magnitude smaller than the belief implies. Where the belief predicts collapse most confidently—the finale—the data show a local maximum in roughly seven cases out of ten.

Decline is nonetheless real in identifiable subsets: long-running series, older series, and a 12.6% minority of ended series whose final season drops materially. That minority is composed disproportionately of widely-watched titles, which offers a parsimonious account of how a belief this durable survives evidence this consistent—the counterexamples are, by construction, the ones nobody talks about.

The effects we can identify are precisely estimated but jointly explain 6% of the variation between series. Whatever determines a show’s trajectory is mostly not its age, its length, or its genre.

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A Full cross-sectional model

OLS of the per-show trend on era, length, ongoing status and 16 non-exclusive genre indicators. $n = 3,234$, $R^2 = 0.0603$, residual SD 0.756. Premiere year is centred; the genre reference category is “not carrying that genre”. Table 3 reports every coefficient.

Table 3: Full cross-sectional model. “Series” counts how many series carry each genre tag; genres are non-exclusive.

Variable	Coefficient	Std. error	t	Series
Intercept	0.2541	0.0537	4.73	—
Premiere year (per year)	0.0048	0.0010	4.81	—
Seasons	−0.0261	0.0038	−6.78	—
Ongoing	−0.0422	0.0387	−1.09	524
Animation	0.2332	0.0418	5.57	680
Action	0.0414	0.0413	1.00	821
Family	0.0254	0.0617	0.41	172
Comedy	−0.0231	0.0389	−0.60	1,365
Drama	−0.0297	0.0359	−0.83	1,733
Crime	−0.0446	0.0381	−1.17	788
History	−0.0506	0.0810	−0.62	101
Mystery	−0.0552	0.0467	−1.18	405
Sci-Fi	−0.0726	0.0745	−0.97	114
Horror	−0.1064	0.0733	−1.45	125
Fantasy	−0.1086	0.0566	−1.92	214
Romance	−0.1428	0.0531	−2.69	254
Thriller	−0.1529	0.0615	−2.49	186
Adventure	−0.1651	0.0440	−3.75	716
Documentary	−0.2345	0.0858	−2.73	101
Biography	−0.2436	0.1083	−2.25	53

B Reproduction

Download the three IMDb tables into `data/raw/`, then run `src/build.py`, the `phase2` scripts, and `phase3_figures.py` in that order. Full instructions are in the repository README.